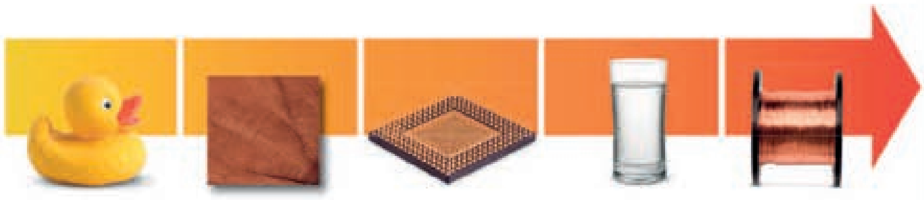
CONDUCTORS AND INSULATORS

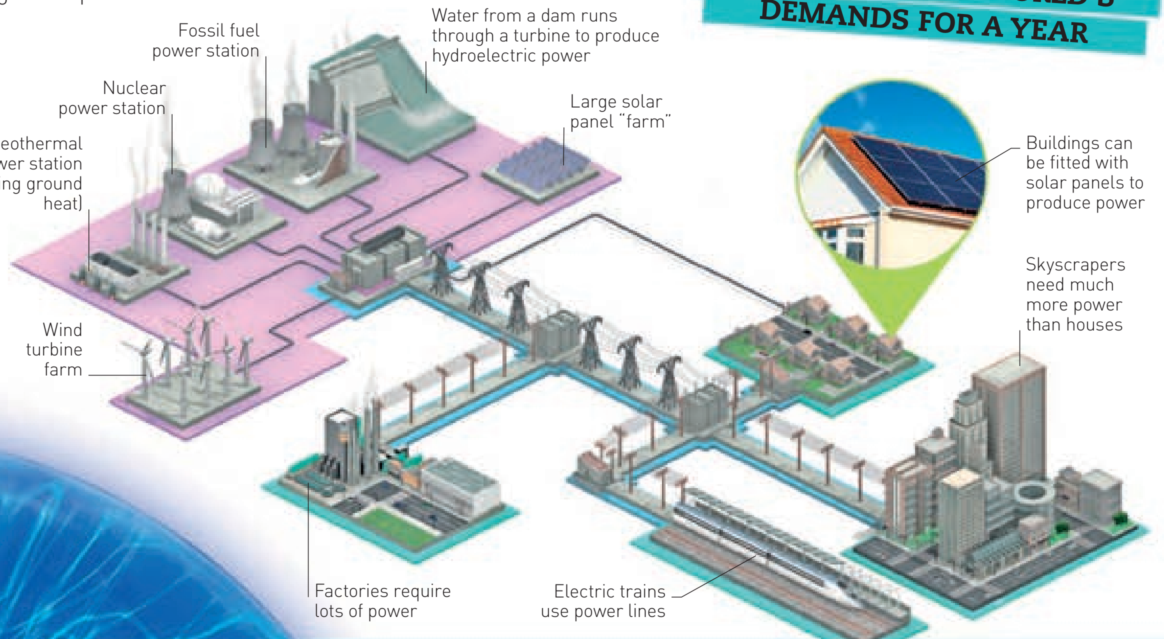
Electricity is a flow of electrons, so materials that do not allow the flow cannot pass along electricity. These are called "insulators". Materials that do allow the flow of electricity are called "conductors".



POWER TO THE HOME

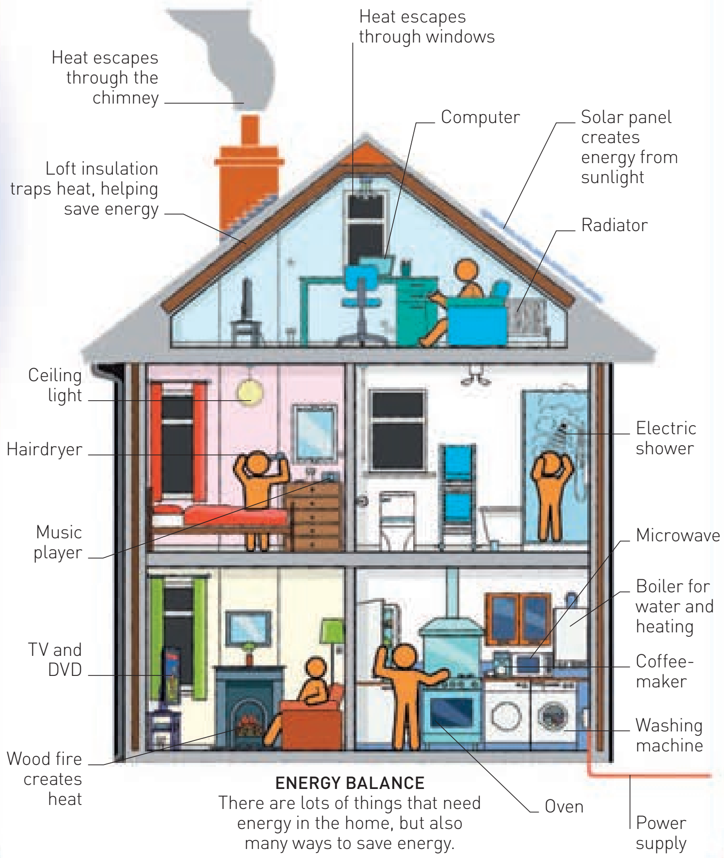
Electricity is produced for homes in several ways, such as burning coal or using nuclear power. The electricity is then fed though sub-stations to individual houses. Some houses also produce their own power through solar panels.

ONE MINUTE OF SUNLIGHT PROVIDES ENOUGH ENERGY TO SATISFY THE WORLD'S DEMANDS FOR A YEAR

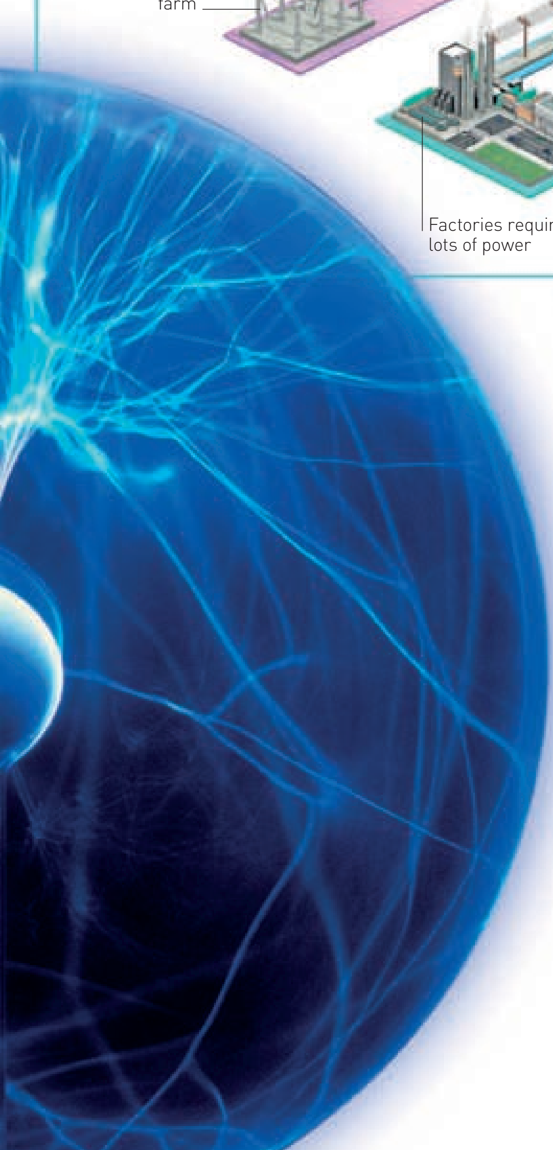


ELECTRICITY AT HOME

We use electricity at home from the moment we get up (perhaps switching on a light or using an electric toothbrush), to when we go to bed. Homes need energy for heat, light, cooking, and washing machines, as well as lots of personal items, such as hairdryers and mobile phones.



ENERGY BALANCE
There are lots of things that need energy in the home, but also many ways to save energy.



PLASMA SPHERE
The streams of plasma here are created by the release of static electricity, which flows as a current from the centre to the edge of the glass sphere.

VOLTAGE

Voltage is a kind of force that makes electricity move through a wire. The bigger the voltage, the more current will shoot through the wire. Bigger voltages and currents deliver more electrical power, but they are also more dangerous.

PYLONS
These hold up overhead lines that carry electricity across long distances. The largest ones use 400,000-volt cables. Cables on wooden poles use 400–11,000 volts.

ELECTRIC TRAIN CABLES
Trains take power from cables above them. One train needs less than 1,000 volts, but the cables are about 25,000 volts. This means many trains can use the line at once.

ELECTRICITY AT HOME
Voltage in the home differs from country to country, but generally lies at 110–250 volts. Factories need higher voltages because they have bigger machines.

BATTERY CHARGERS
A laptop or phone charger needs 10–20 volts to charge its battery. Laptops need higher voltages than phones because they have bigger screens and circuits that use more energy.

TORCH BULBS
Bulbs for torches and lamps are rated by the voltage and current needed to operate them. The standard AA, C, and D batteries all deliver 1.5 volts each.

PIONEERS

Electricity has been around forever, because it exists naturally in the world. However, some people were important in finding out how to harness its power.

BENJAMIN FRANKLIN (1706–90)
Franklin discovered that lightning is electricity, and that there are positive and negative charges.

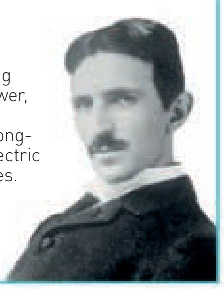
ALESSANDRO VOLTA (1745–1827)
A professor of experimental physics, Volta invented the first battery, called the Voltaic Pile.

GEORG SIMON OHM (1789–1854)
Ohm discovered electrical resistance. The unit of resistance – ohm – is named after him.

MICHAEL FARADAY (1791–1867)
Faraday discovered that if you move a magnet near wire, the wire becomes electrified. This is known as electromagnetic induction.

THOMAS ALVA EDISON (1847–1931)
Edison built the first electric power stations and invented the lightbulb, sound recorder (phonograph), and movie camera.

NIKOLA TESLA (1856–1943)
Tesla discovered alternating currents, hydroelectric power, radio waves, and radar. He invented transformers, a long-distance power system, electric motors, and X-ray machines.



NIKOLA TESLA